

Algebra II with Trigonometry

Chapter 5.5

Olympiad Problems

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Problems

1. In triangle ABC, angle A is twice angle B and side a is twice side b. What is the value of angle C?
2. In triangle PQR, $PQ = 5$, $QR = 7$, and angle $Q = 60^\circ$. Without using a calculator, determine whether two distinct triangles can be constructed with these parameters. Justify mentally.
3. Let triangle XYZ have sides $x = 8$, $y = 15$, and angle $X = 30^\circ$. Show mentally whether a triangle exists, and if so, whether more than one triangle is possible (the ambiguous case).
4. Triangle DEF has angles D, E, F such that $\sin D + \sin E = \sin F$. If sides d, e, f are opposite these angles respectively, what can you say about the ratios $d : e : f$?
5. In triangle LMN, $LM = 10$, $MN = 7$, and angle $N = 90^\circ$. Find $\sin L + \sin M$ using only properties of sines in a triangle.
6. Triangle ABC has sides a, b, c opposite angles A, B, C respectively, with $a = \sqrt{3}$, $b = 1$, and angle $C = 120^\circ$. Find, without explicit calculation, whether triangle ABC is isosceles, scalene, or impossible.